Week 4-5 Technical Report: Attrition Prediction Pipeline

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1. Project Goal

The objective of this Capstone is to develop a predictive model for employee attrition that prioritizes interpretability alongside accuracy. High turnover rates in critical sectors (such as the mental health field identified by Wu & Fukui) are often crippling in that they create a negative cycle of increased workload and diminished morale for retained staff [1]. By leveraging HR data this project aims to fashion a predictive model with sufficient explanatory power (gathered from dataset features that represent various turnover factors) that is conducive to timely interventional actions on behalf of the employing organization [2].

2. Methods: The MARS Framework

The guiding framework in interpreting model results comes to us from the MARS model that was introduced in Week 2's literature review. Interpreting results under the MARS framework allows for explanations as to "why" turnover factors are present and thus serves as a veritable complement to mathematically computed results, which tell us "where" stressors and turnover factors are concentrated in the form of feature importances and model coefficients. As a recap, the four prongs of the MARS model are listed below with a statement on their relevance to the task at hand:

- **Motivation (M):** These are the well-documented internal engines of behavior, particularly turnover intent. To measure this, `JobSatisfaction`, `TrainingTimesLastYear`, and `StockOptionLevel` columns have been selected. Should they rank high in importance, they would point to a morale deficit contributing to turnover.
- **Ability (A):** This prong is a proxy for an employee's capabilities and aptitude in their employed role. Mapped here are `PerformanceRating`, `Education`, `EducationField`, and `YearsAtCompany`. A spike in turnover linked to these variables suggests a skills mismatch or lack of career development (opportunities or interest).

- **Role Perceptions (R):** This defines the degree to which the employee understands their job or role and what is consequently expected of them in those capacities. Here `Department`, `JobLevel`, `YearsWithCurrManager`, and `YearsInCurrentRole` will be tested as proxies for role perception.
- **Situational Factors (S):** These are the external factors that influence turnover that lie beyond the scope of the employee's role and decision-making power. This is inclusive of factors ranging from mergers and acquisitions, company profitability, and regulatory environments to more individualistic factors like workspace layout and commute distances. This is simultaneously the most critical category for intervention and the most challenging to measure and conceptualize as variables. I have isolated `DistanceFromHome`, `OverTime`, and `BusinessTravel` to test if logistical burnout is the primary driver of attrition, and have included `EnvironmentSatisfaction` as a stand-in variable for the aforementioned external factors.

2.1 The Comparative Panel

To ensure the final results are both benchmarked and robust, and to mitigate the risk of overreliance on a single technique, a panel of five classifier algorithms will be used and compared against an untuned benchmark model:

- **Model A: Standard Logistic Regression (The Benchmark).** This model assumes a linear relationship between predictive features and turnover odds without correction via penalties. Selecting this model allows us to validate Wu's observations about Multicollinearity in HR data; specifically, we test if the model fails to produce stable coefficients in the presence of correlated features [1].
- **Model B: Penalized Logistic Regression (Ridge Method).** Here, the remedial step articulated by Wu & Fukui [1] for multicollinearity can be tested with the introduction of the L2 penalty term. This L2 penalty term allows for the shrinking of noise from correlated variables (like `YearsAtCompany` vs `YearsInCurrentRole`) rather than letting their interactions compromise the integrity of the model.
- **Model C: Linear SVM (Support Vector Machine).** The choice of this model allows us to approach turnover from a novel angle. Whereas Logistic Regression deals with probabilities, and Decision Trees with conditional logic, Linear SVMs tackle the turnover problem by using geometry (i.e., constructing a hyperplane to serve as a "physical" boundary between stayers and leavers). Although more sophisticated kernels such as the Radial Basis Function (RBF) are available, confining our SVM to the Linear Kernel allows us to pay fidelity to the project's core objective of interpretability [3].
- **Model D: Structural Decision Tree (The Mantovani Protocol).** This model is present in the classification ensemble as it is the only model on the panel to offer logical rules instead of coefficients. As such, Decision Trees are some of the most workable models in conveying model results to stakeholders in a manner most propitious to their intuitive understanding, following the protocols of Mantovani et al. [4].

- **Model E: K-Nearest Neighbors (kNN Algorithm).** This model abandons coefficients entirely in favor of Euclidean Distance. To ensure the distance calculation is not biased by variable magnitude (e.g., `DailyRate` vs. `JobSatisfaction`), the input data will be strictly scaled, as KNN algorithms have known vulnerabilities when encountering unscaled data. The number of nearest neighbors used for prediction (k) is obtained empirically and varies by dataset structure.

2.2 Methodological Steps

- **Input:** Raw IBM HR Dataset D (1,470 rows, 35 features obtained from Kaggle).
- **Output:** Binary Classification Model M, Variable Importance Vector V.

Step 1: Preprocessing

- **Drop Non-Informative Columns:** $\{EmployeeCount, StandardHours, Over18\}$. As these columns contain identical values for all records in the dataset, they convey no predictive signal and are removed to prevent singular matrix errors.
- **Encode Categorical Variables (X_{cat}):** Variables such as `BusinessTravel` are transformed using One-Hot Encoding. Machine Learning libraries (like sklearn in Python) necessitate inputs as numerical arrays, requiring this conversion from text strings to binary flags.
- **Scale Numerical Variables (X_{num}):** Variables such as `DailyRate` are normalized using standard MinMax Scaling to prevent magnitude bias. Absent this step, variables like `MonthlyRate` (values in the thousands) would mathematically dominate variables like `StockOptionLevel` (values 0-3), distorting the distance calculations in Model E (kNN) and Model C (SVM).

Step 2: Model A (Standard Logistic Regression)

- **Implementation:** Logistic Regression with No Penalty (L1 or L2).
- **Justification:** As standard logistic regression typically fails on correlated HR data, this unregularized iteration serves strictly as a Reference Model to benchmark the sophisticated models against.

Step 3: Model B (Regularized Logistic Regression)

- **Implementation:** Logistic Regression with Elastic Net/Ridge penalty.
- **Justification:** Following Wu[1], I apply an L2 penalty to shrink coefficients rather than remove them, explicitly pursuing a more stable model than the benchmark (Model A).

Step 4: Model C (Linear SVM)

- **Implementation:** Support Vector Classifier with a Linear Kernel.

- **Justification:** As stated in Section 2.1, this model allows us to seek a geometric solution to the problem, defining a decision boundary that probabilistic models may miss.

Step 5: Model D (Decision Tree)

- **Implementation:** CART with structural tuning.
- **Justification:** Mantovani [4] proves that default tree depth leads to overfitting. I explicitly restrict tree growth using `max_depth` and `min_samples_leaf` to ensure the generated rules remain generalizable.
- **Hyperparameter Grid:** `max_depth` $\in \{3, 5, 10\}$, `min_samples_leaf` $\in \{10, 20, 50\}$.

Step 6: Model E (kNN Classification)

- **Implementation:** K-Nearest Neighbors Classifier.
- **Justification:** kNN models are particularly impactful in the delivery of model interpretations to stakeholders. It allows for case-based reasoning (e.g., explaining that an employee is at risk because they share similarities with 5 specific former employees who terminated).
- **Hyperparameter Grid:** k (number of neighbors) is the critical hyperparameter and will be estimated via an Elbow Diagram.

Step 7: Evaluation Strategy

The evaluation will be grounded on an array of diagnostic techniques discussed below that will be applied to all the above models where applicable:

- **The Case for Recall:** The use of Accuracy as a primary metric in attrition studies often produces results that are nebulos or skewed. A turnover classification model simply predicting that "every employee stays" could achieve an accuracy rate of ~84% due to the high class imbalance (No-Information Rate). Therefore, Recall (the percentage of actual leavers a model correctly identifies) is the pivotal metric under the auspices of Workforce Management.
- **F1-Score:** As Recall and Precision are often inversely related, the inclusion of the F1-Score offers a balanced metric for the model's overall capabilities.
- **Elbow Diagram:** A diagnostic solution particular to kNN algorithms, the Elbow Diagram plots Model Error (Y-axis) against the number of neighbors k (X-axis). The "Elbow" occurs where the marginal gain in predictive power diminishes. The optimal k is selected at this inflection point (e.g. if the error rate stabilizes at $k = 4$, then 4 neighbors are selected to balance complexity and bias).
- **Variable Importance Measures (VIMs):** In computing this metric, relative scores are assigned to all model features, representing their proportional importance with respect to the most influential variable. These scores assist in isolating Global turnover factors that can be primed for HR intervention (e.g., identifying if `OverTime` is the dominant driver across the entire organization).

- **Shapley Values (SHAP):** This is a Localized metric used to diagnose why a specific individual was flagged. SHAP values are delivered as case-wise statistics that delineate the contribution of a particular predictor (eg, `DistanceFromHome`) taken against the baseline. Due to the granular nature of these metrics, we will confine their use to specific edge cases, just as Wu & Fukui [1] did.

2.3 Key Mathematical Formulations

To ensure statistical validity and reproducibility, the analysis relies on the following mathematical frameworks governing the classifiers and evaluation metrics:

A. The Logit Function (Probability Estimation)

To generate the probability scores required for the SHAP diagnostic (identifying "edge cases"), Models A and B utilize the sigmoid function to map linear predictions into a probability space between 0 and 1:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x_1+\ldots+\beta_n x_n)}}$$

While standard regression models output raw values in their predictions, Logistic Regression—which is grounded on the Logit function—returns probabilities of each record being classified as the positive class using a threshold. For example, employee attrition models return the probability of an employee terminating, and final predictions are determined if this probability clears a specific threshold (e.g. > 0.5). Probabilities are particularly influential in the HR space as they allow the user to discern which employees present more risk than others in terms of propensity to depart the organization.

Note: The β terms represent variable coefficients and Y represents the dependent variable of concern.

B. Penalized Loss Function (Ridge/L2)

To handle the multicollinearity identified by Wu & Fukui, Model B minimizes a loss function that includes a penalty term for coefficient magnitude, strictly constraining variance [1]:

$$J(\theta) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(h_\theta(x_i)) + (1 - y_i)\log(1 - h_\theta(x_i))] + \lambda\sum_{j=1}^{p}\theta_j^2$$

Due to the variability of HR data and its susceptibility to stochastic trends, standard logistic regression models tend to overfit the training data by erroneously accommodating noise into the model. Regularization via an L2 parameter allows us to mitigate the effects of collinearity by penalizing correlated variables. The penalty cost (or L2 term) is denoted by λ in the above formula.

C. Linear Kernel SVM (Soft Margin Optimization)

Model C constructs a decision boundary by solving a constrained optimization problem. Unlike the hard-margin ideal, this Soft Margin formulation introduces slack variables (ξ) to tolerate some misclassification, preventing the model from over-fitting to outliers:

Minimize:

$$\min_{w,b,\xi} \frac{1}{2}||w||^2 + C \sum_{i=1}^{n} \xi_i$$

Subject to:

$$y_i(w \cdot x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

The formula above attempts to mediate between two contradictory goals. The first half ($||w||^2$) seeks to maximize the width of the hyperplane separating stayers and leavers. The latter half ($C \sum \xi_i$) serves as a penalty term for each mistake the model makes. Finally, the slack variable (ξ_i) functions as a bulwark against model rigidity and overfitting by permitting the model to misclassify a limited number of outliers, provided that the interpretability and straightness of the decision boundary are maintained.

D. Euclidean Distance (kNN)

Model E (kNN) relies on geometric proximity rather than coefficients. The similarity between a target employee x and a neighbor y is calculated as the square root of the sum of squared differences:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

The formula above underscores the importance of scaling, as variables with large variations in magnitude would skew Euclidean computations that rely on squared distances. Without scaling, an employee's salary (e.g., 5000) would wrongly be treated as significantly more influential than the number of promotions (e.g., 2), simply due to the numerical difference.

E. Shapley Additive Explanations (SHAP)

To diagnose the *local* contribution of each feature to a specific employee's risk score, I calculate the Shapley value (ϕ), which represents the weighted average marginal contribution of a feature value across all possible feature coalitions:

$$\phi_j(val) = \sum_{S \subseteq \{x_1, \ldots, x_p\} \setminus \{x_j\}} \frac{|S|!(p - |S| - 1)!}{p!} (val(S \cup \{x_j\}) - val(S))$$

Whereas VIM scores rank variables globally as per their predictive power, SHAP zeroes in on individual predictions. In an attrition model, the organization's overall turnover functions as a baseline; SHAP illuminates how much a particular employee's attributes influence their specific risk relative to that baseline. This allows us to replicate Wu & Fukui's juxtaposition of the highest-risk candidates against the lowest-risk ones [1].

F. Performance Metrics

Per the evaluation strategy, the model is graded on its ability to capture true leavers (Recall) and the harmonic balance of its precision (F1-Score):

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Recall the first measure above is the proportion of True Positives to the sum of True Positives and False Negatives. It functions as a safety net to estimate how many flight risks were caught versus how many were missed. F1-Score the measure below is included to ground our interpretation from a myopic focus on recall. By balancing precision and recall, F1-Scores ensure that models aren't erroneously rewarded for achieving high accuracy if all they are doing is classifying a preponderance of records to the positive class.

3. Pre-Processing Data

This section on the report focuses on the pre processing of the data at hand, Throught this document only the outputs from the code are displayed to guide the reader through the workflow:

3.1 Data Loading

At first, the data is loaded and the aforementioned MARS specific columns from Section 2 are retained while others are dropped, Below is a rendering of the shapes of the IBM dataset before and after feature pruning, after this the first five records are observed to glean which columns must be transformed:

```
Original Data Shape: (1470, 35)
MARS Filtered Shape: (1470, 16)

Missing Values Check:
0

First 5 Rows:
```

	Attrition	Education	EducationField	Department	OverTime	BusinessTravel	Env
1	No	1	Life Sciences	Research & Development	No	Travel_Frequently	3
2	Yes	2	Other	Research & Development	Yes	Travel_Rarely	4
3	No	4	Life Sciences	Research & Development	Yes	Travel_Frequently	4
4	No	1	Medical	Research & Development	No	Travel_Rarely	1

3.2 Categorical Variable Encoding

From the dataset preview above, it is apparanet that the target `Attrition` column and the `OverTime` column must be binary encoded.

Below is an example of the `Attrition` column being appropriately encoded:

```
 The following dictionary is used to refactor the Attrition column:

 {'Yes': 1, 'No': 0}

 These are the contents of the Attrition column after refactoring:
```

```
count    1470.000000
mean        0.161224
std         0.367863
min         0.000000
25%         0.000000
50%         0.000000
75%         0.000000
max         1.000000
Name: Attrition, dtype: float64
```

Similarly the `BusinessTravel` column is categorical with low cardinality and can therefore be encoded manually as the label in this column are sequential in terms of magnitude:

```
 The following dictionary is used to refactor the BusinessTravel column:

 {'Travel_Rarely': 1, 'Travel_Frequently': 2, 'Non-Travel': 0}

 These are the contents of the BusinessTravel column after refactoring:
```

```
BusinessTravel
1    1043
2     277
0     150
Name: count, dtype: int64
```

3.3 One Hot Encoding & X/Y Split

As employee educational backgrounds and departmental placements are subject to signficant variability, the `EducationField` and `Department` must be treated differently as they are natural high cardinality columns. The **get_dummies()** from the **Pandas** library will be used to convert these columns into series' of binary flag columns for each class label.

A Train test split (a commonplace practice in Machine Learning to constrain overfit) will be employed, 85% of available data will be used to train each model and 15% of the data that remains will be retained as a test set.

Below, the aforementioned columns previewed before and after the application of the **get_dummies()** function

```
 EducationField and Department columns prior to get_dummies :

   EducationField              Department
1  Life Sciences  Research & Development
2          Other  Research & Development
3  Life Sciences  Research & Development
4        Medical  Research & Development

 EducationField and Department columns after get_dummies :

    EducationField_Life Sciences  EducationField_Medical  \
1                           True                   False
2                          False                   False
3                           True                   False
4                          False                    True

   EducationField_Other  Department_Research & Development
1                 False                               True
2                  True                               True
3                 False                               True
4                 False                               True
```

3.4 Feature Scaling & Class Weights

Below we observe that the target variable is experiencing a class imbalance issue that can be remedied by utilizing class wieghts in classification models to balance the classes for prediction. This is done using the `compute_class_weight` function from the `sklearn.utils.class_weight` Python module:

```
 Class distribution of values in Target (Attrition) Column :

 Attrition
0    1233
1     237
Name: count, dtype: int64
The resulting set of class weights to remedy imbalance is : {0: 0.5961070559610706, 1: 3.101
```

Therefore it appears that to correctly weight the model for analysis we must assign the positive class (`Attrition` = 1) six times the priority of the negative class.

4 Model Construction & Interpretation

With the dataset having been tidied and prepared for analysis in the previous section, this section will hone in on the construction of models and the insights that diagnostics performed on them can tell us. As such, each Model will be reviewed with following sequence:

Algorithm: Model Training & Interpretation Pipeline

```
FOR EACH Model_Type IN {Logistic, Ridge, SVM, Tree, kNN}:
    # 1. Hyperparameter Optimization
    Define Grid_Space (G)
    Best_Model = RandomizedSearchCV(Model_Type, Grid_Space, X_train)

    # 2. Prediction
    y_pred = Best_Model.predict(X_test)

    # 3. Evaluation
    Generate Confusion_Matrix(y_test, y_pred)
    Calculate Recall, F1_Score

    # 4. Global Interpretation (Feature Importance)
    IF Model has Coefficients:
```

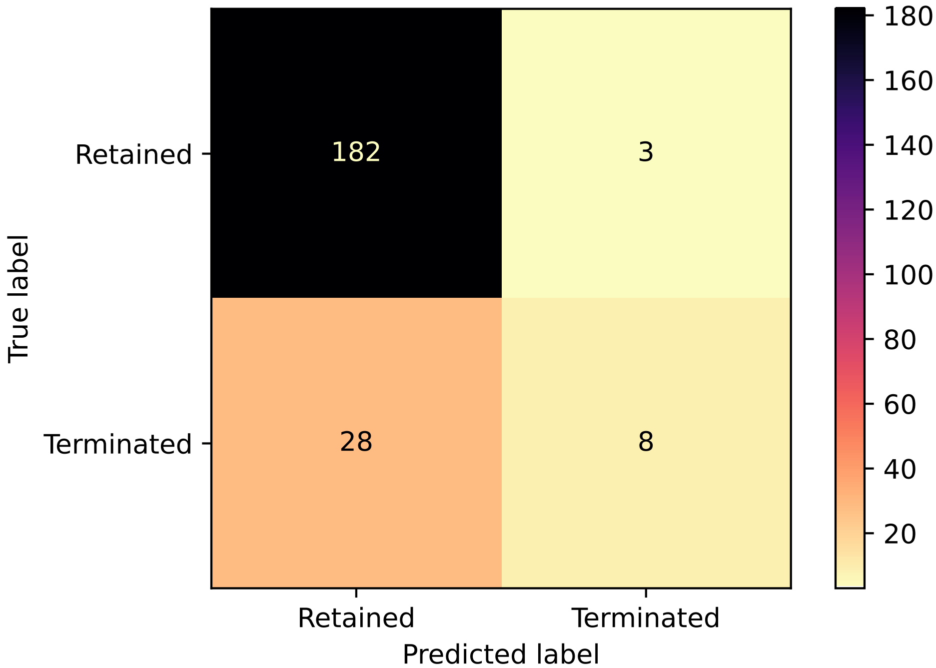
```
        Extract Beta_Values -> Sort -> Plot(Top_10)
    ELSE (Tree/kNN):
        Extract Gini_Importance OR Permutation_Importance -> Plot(Top_10)

    # 5. Local Interpretation (Edge Cases)
    IF Model == Logistic OR SVM:
        Identify Edge_Case_Indices (Highest/Lowest Probability)
        Calculate SHAP_Values(Best_Model, Edge_Case) -> Plot(Waterfall)
END FOR
```

4.1 Standard Logistic Regression (Model A)

We embark on our analysis with the Benchmark model. Per Wu & Fukui [1], it is not expected that this model will conform to the irregularities of HR data:

Below is the Classification Summary Report for Model A:

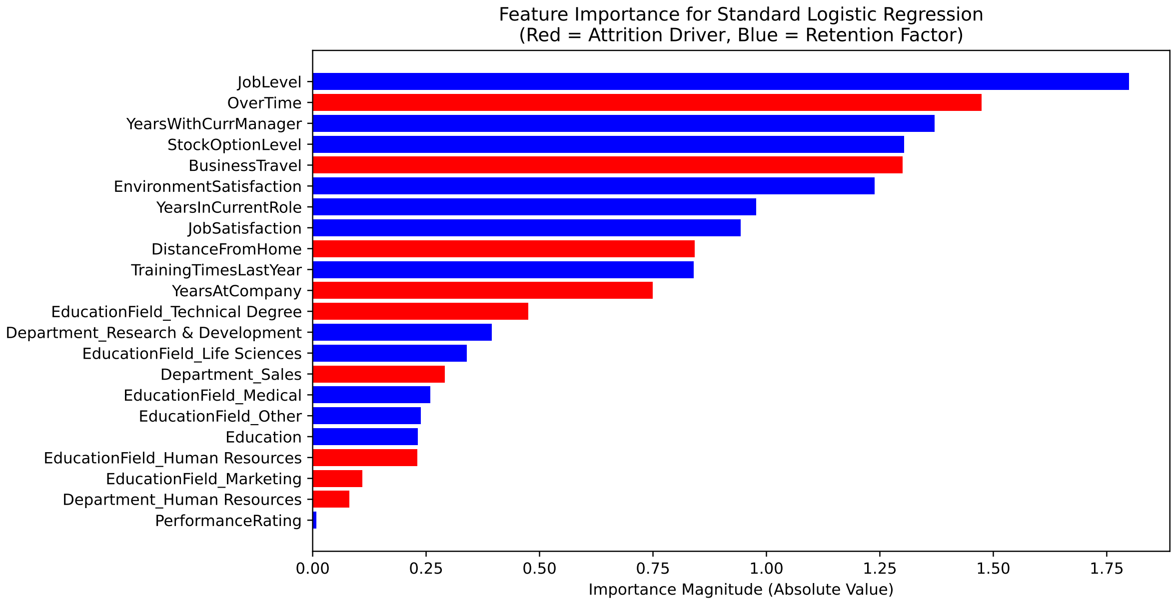
```
The confusion matrix for the Standard Logistic Regression model is :
None
The classification reprort for the Standard Logistic Regression model is :
              precision    recall  f1-score   support

           0       0.87      0.98      0.92       185
           1       0.73      0.22      0.34        36

    accuracy                           0.86       221
   macro avg       0.80      0.60      0.63       221
weighted avg       0.84      0.86      0.83       221
```



With the summary above it is apparent that the classification model above fell into the accuracy trap referenced in Section 2. The overall model has achieved 83% percent accuracy but has done so by predicting nearly all candidates in the test set (210 of 221) as having been retained. We further observe that with a recall of 22% for the positive class, 4 in 5 leavers are incorrectly predicted as retained which serve to compromise the utility of this model.

4.1.1 Feature Importance & SHAPs Analysis

VIMs scores are obtained and plotted for the Standard Logistic Regression Model Below:

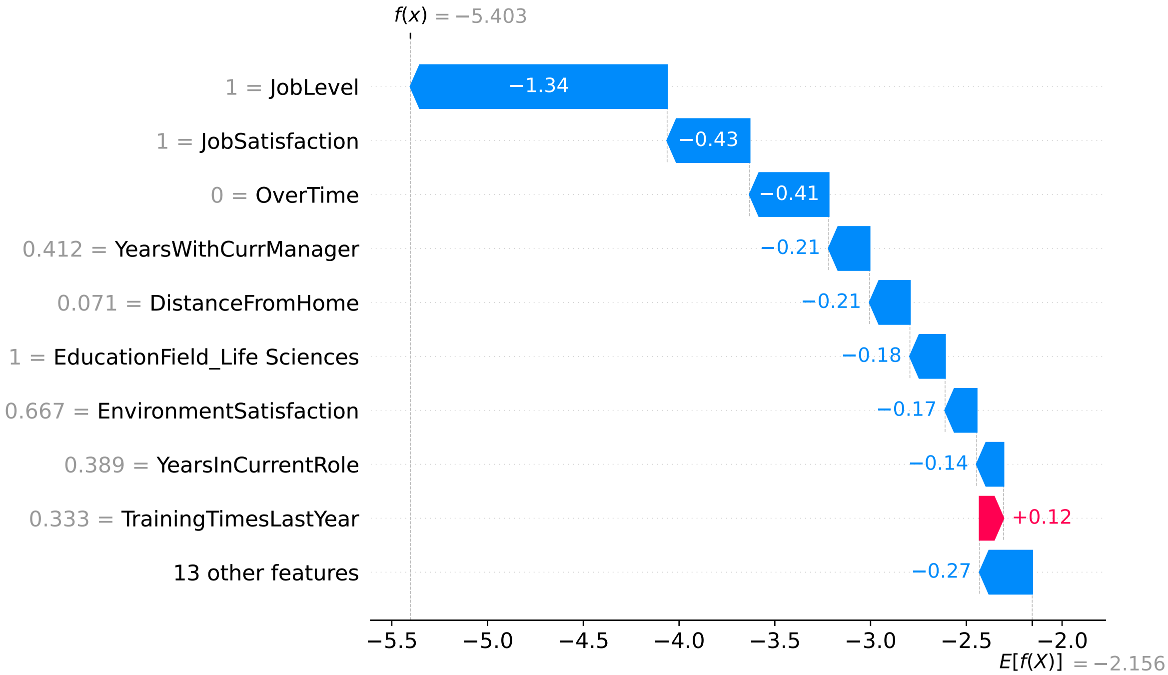


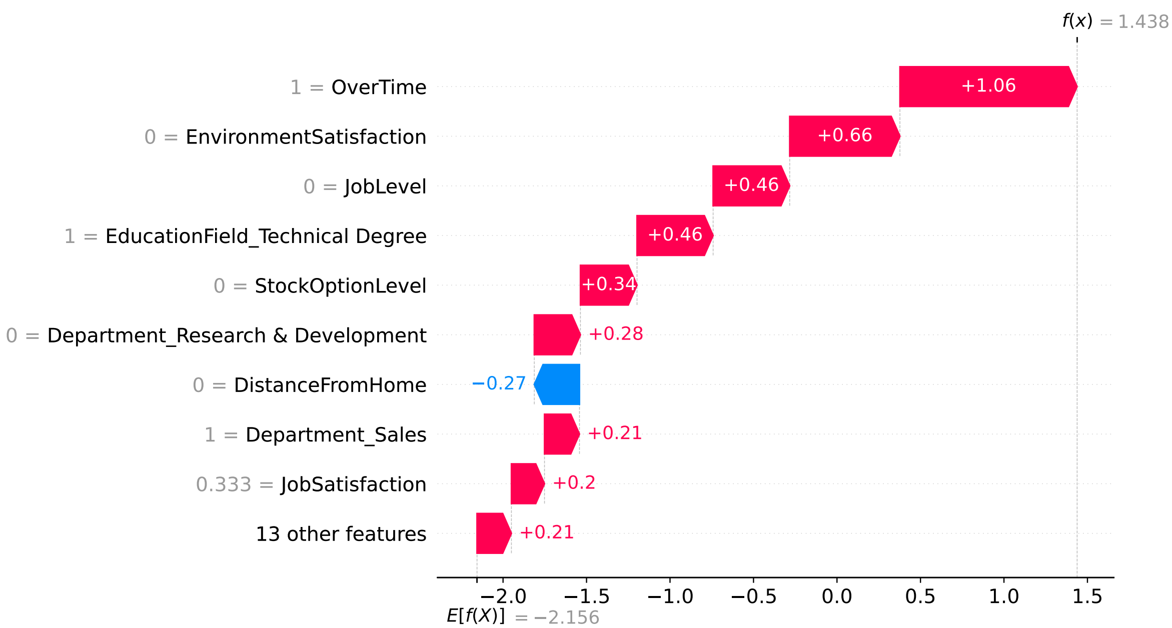

```
-> Added 22 features from Standard Logistic Regression to global tracker.
```

The summary above imparts to us that Overtime status(`OverTime`) and Travel requirements(`BusinessTravel`) are among the chief motivators of turnover indicating Situational factors being the key reason that people terminate per the Benchmark Model. Conversley, Seniority (`JobLevel`) and Familiarity (`YearsWithCurrManager`) influence retention pointing to positive role perception and self concept among particpants being crucial to their longevity.

Below we isolate the highest risk and lowest risk cases (using probabilites obtained from the logit function of the Logistic Regression Model) by constructing SHAPs plots to determine the magnitude and direction of turnover factors on each studied employee's decision to leave or remain.

```
Lowest Risk Index: 140 (Prob: 0.0045)
Highest Risk Index: 47 (Prob: 0.8082)

SHAP Explanation for Lowest Risk Employee (Index 140):

SHAP Explanation for Highest Risk Employee (Index 47):
```





Our SHAPs plots for Model A inform us that the lowest risk employee is primarily being influenced to stay due to their Role Perception or seniority (The 1=JobLevel in the X axis of the of the plot for the lowest risk employee(Index 140) indicates the highest job level as the `JobLevel` variable was scaled between 0 and 1)

Maximal Job satisfaction (`JobSatisfaction` = 1 on a scale of 0 to 1) and exemption from overtime status (`OverTime` = 0) are other influencers of this employee's decision to stay which are intutively sound signals from the model.

Conversely the SHAPs plots tell us that our highest risk employee (Index 47 predicted with an 80% chance of termination) is motivated to depart in part due to entry level status (`JobLevel` = 0 on a scale from 0 and 1). Stiational Factors like overtime requirements and dissatisfaction with the environment are apparent as other potential stressors to this employee.

4.2 Regularized Logistic Regression (Model B)

We now apply regularization to our Logistic Regression model and will tune the following Hyperparameters using RandomizedSearchCV:

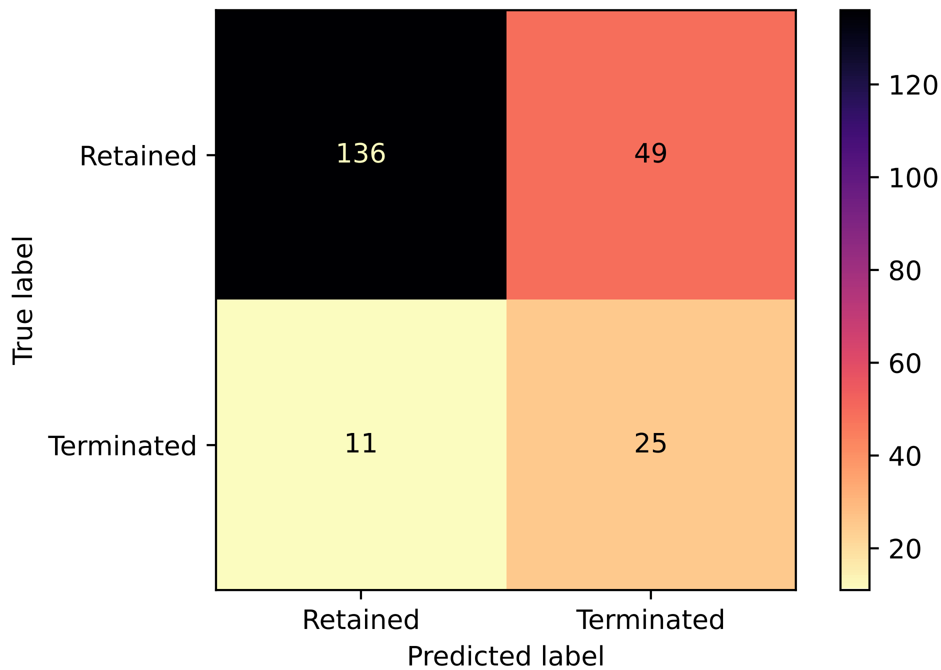
- **C**: This is the inverse of the regularization parameter(L2), where values ranging from 0.001 to 1000 will be tested
- **solver**: This is the optimization algorithm used (here we will offer the grid choices between *newton-cg* and *saga* which are both well suited to L2 regularization and small datasets)
- **class weight**: The grid will be offered a choice between using no corrective weights at all or the class weights computed in Section 3.4

We will use f1 scores as the optimization metric for parameter tuning below:

```
Optimal Parameters for the Regularized Logistic Regression model are : {'solver': 'newton-cg
The test accuracy of the Regularized Logisitic Regression model with optimal parameters is :
```

As the Random Search grid has settled on a model variant that utilizes the corrective class weights obtained earlier, we inspect the Classification report for this enhanced model (Model B) below:

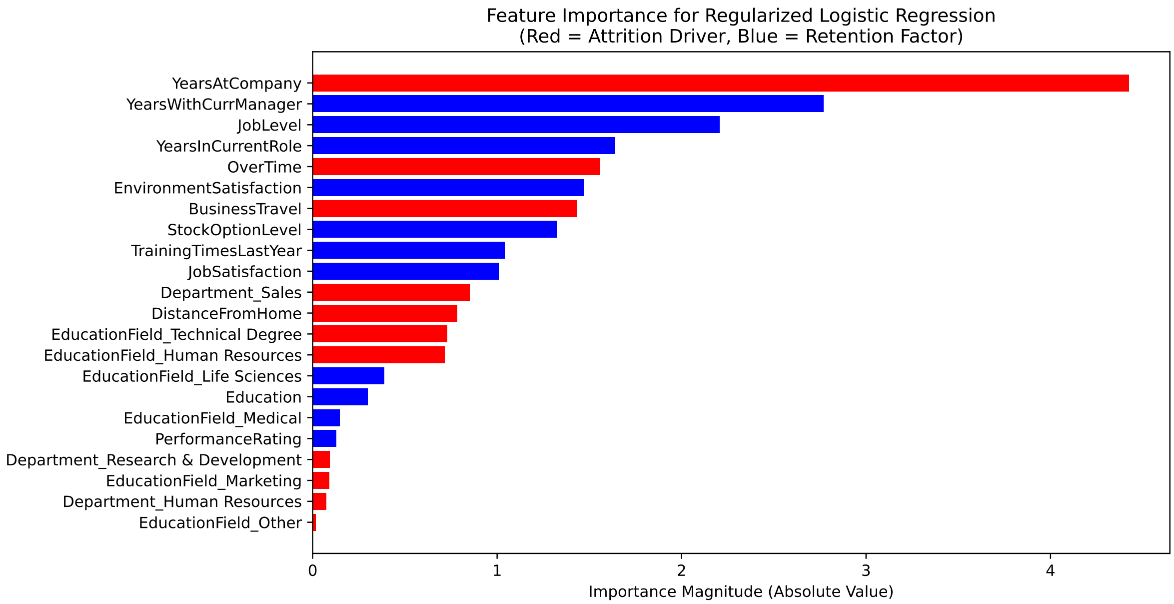
```
The confusion matrix for the Regularized Logistic Regression model is :
None
The classification reprort for the Regularized Logistic Regression model is :
              precision    recall  f1-score   support

           0       0.93      0.74      0.82       185
           1       0.34      0.69      0.45        36

    accuracy                           0.73       221
   macro avg       0.63      0.71      0.64       221
weighted avg       0.83      0.73      0.76       221
```



Based on the report above Model B maintains a slightly lower accuracy rate with Model A (0.76 vs O.82) but is the more functionally sound model with a Recall rate of 0.69 for the positive class compared to the 0.22 for model A. However, as tuning precision and recall often involve tradeoffs, this has led to a significant drop in precision with respect to model A.

However, from an HR perspective this would be regarded as tolerable, as the 0.34 Precision model B produces more false positives which often materialize as diaglogue with the employee centered around incentivizing retention. Lower rates of recall however represent lost opportunities to stem attrition and Model B boasts a signicant performance improvement in keeping this risk in check by identifying 3 times as many leavers as Model A (69% in Model B vs 22% in Model A) making it the more viable model between the pair despite lower accuracy.

4.2.1 Feature Importance & SHAPs Analysis



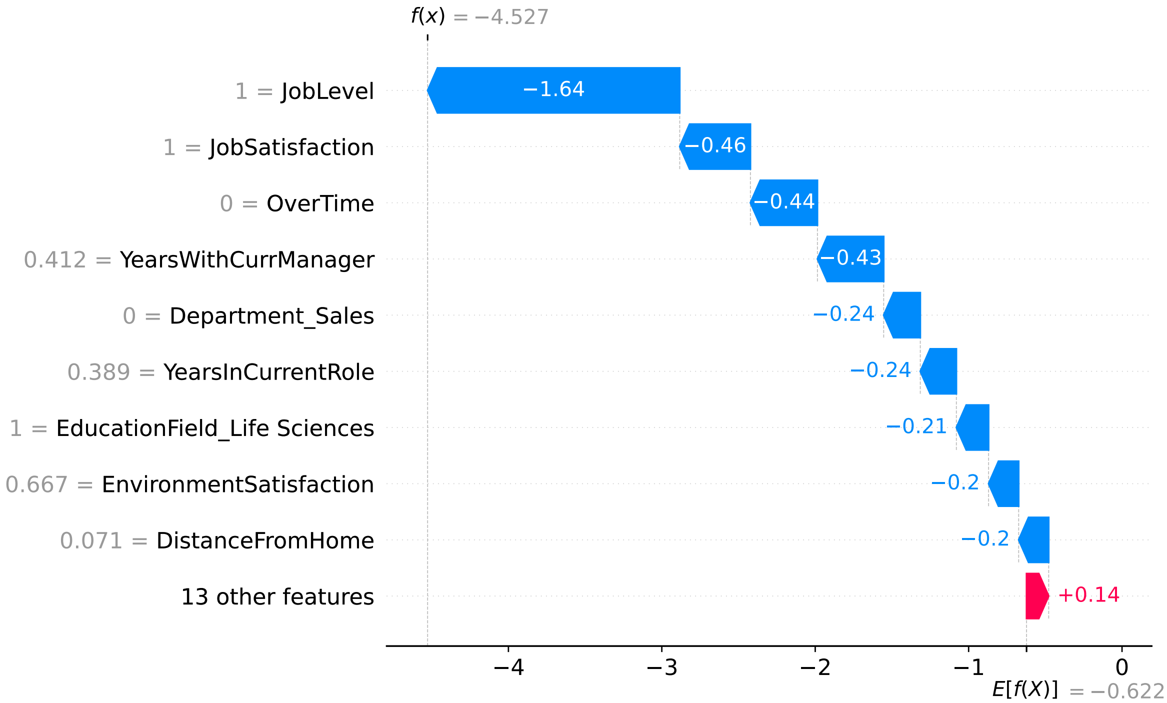

```
-> Added 22 features from Regularized Logistic Regression to global tracker.
```

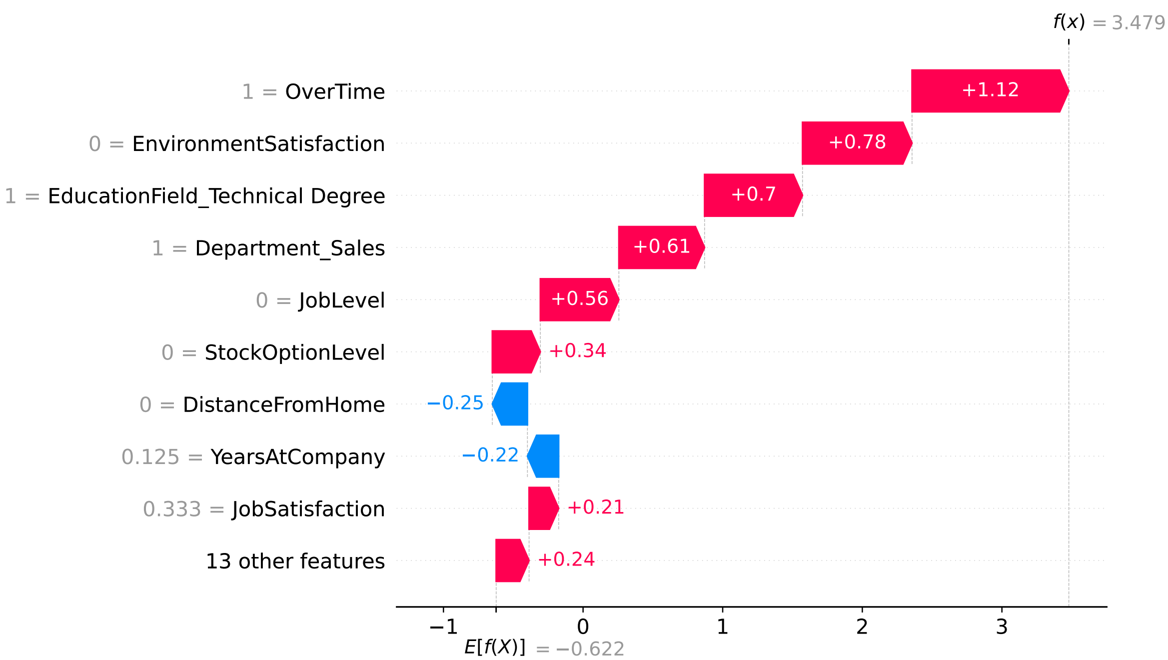
Gleaning the feature importance plot, almost readily noticeable is the emergence of the stagnation signal from model output data. The most dramatic shift between Model A and Model B occurred in the `YearsAtCompany` variable which was priortized by the standard model (Model A) as a middling factor has shot up to the prime reason for attrition.

Taken in tandem with the most potent retention factor `YearsWithCurrManager`, it becomes apparent to us that Tenure cannot soley be relied upon as a proxy for organizational loyalty. Model B suggests that tenured employee's who don't form a deep connection their managers will experience a "rotting on the vine" effect and attrite. This nuance was completely invisible in the benchmark model thereby enshrining the necessity of techniques like the L2 penalty in decoding complex HR dynamics.

We now turn to SHAPs for more localized insights on the highest and lowest risk employees:

```
Model B - Lowest Risk Index: 140 (Prob: 0.0107)
Model B - Highest Risk Index: 47 (Prob: 0.9701)

SHAP Explanation for Model B's Safest Employee (Index 140):

SHAP Explanation for Model B's Riskiest Employee (Index 47):
```





From the plots above we observe that both Models thus far the standard and regularized logistic regression models have identifed the exact same employees as edge cases (employee #47 as highest risk and employee# 140 as lowest risk).

However the effects of regularization and the penalty term are best observed here by the overall confidence scores recorded in both models for employee #47 (where $f(x) = 3.47$ (97% probability of termination) for Model A and)$f(x) = 1.48$ (80% probability of termination) for Model A. This informs us that while Model A was suggesting the departure of employee #47, Model B was screaming it pointing to tuned model being "radicalized" by its penalty term.

Also noteworthy with employee #47 is tenure (YearsAtCompany) being a retention factor for them while being an attrition factor for the rest of the organization. When `OverTime` the number 1 attrition driver for employee #47 is considered we can infer that this employee was a new hire burnout who was overworked and dissatisfied with their environment.

4.3 Linear SVM (Model C)

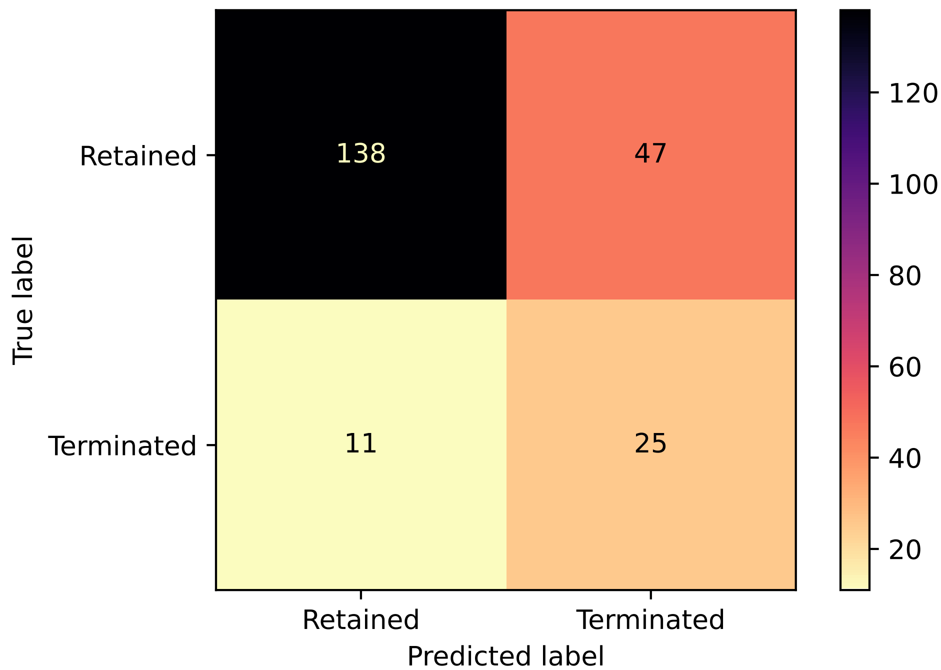
We now apply regularization to our Logistic Regression model and will tune the following Hyperparameters using RandomizedSearchCV:

- **C**: This is the inverse of the regularization parameter(L2), where values ranging from 0.001 to 1000 will be tested

```
Optimal Parameters for the SVM model are : {'class_weight': {0: 0.5961070559610706, 1: 3.101
The test accuracy of the SVM model with optimal parameters is : 0.47540383141762454
```

Having arrived at the optimal configuration of parameters for our Linear SVM Kernel, we train the model and produce the classification report below:

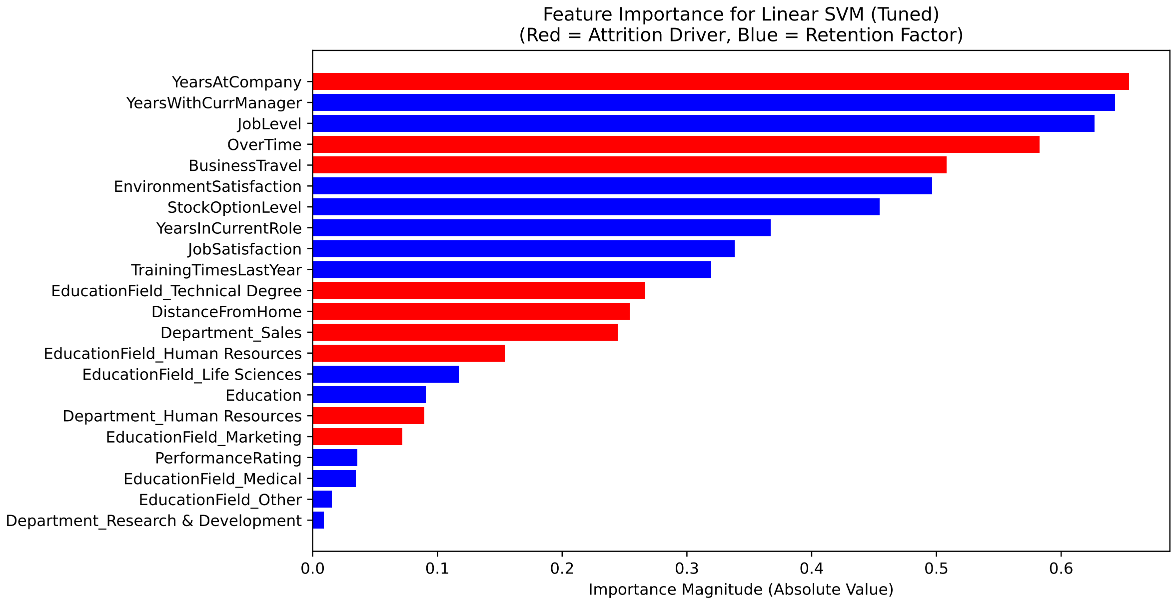
```
The confusion matrix for the Linear SVM (Tuned) model is :
None
The classification reprort for the Linear SVM (Tuned) model is :
              precision    recall  f1-score   support

           0       0.93      0.75      0.83       185
           1       0.35      0.69      0.46        36

    accuracy                           0.74       221
   macro avg       0.64      0.72      0.64       221
weighted avg       0.83      0.74      0.77       221
```



This classification report while bearing a simiarity to Model B's classification report has traded off a small amount of recall (0.71 in Model B vs 0.69 in Model C) for precision (O.33 in Model B vs 0.35 Model C) which indicates that the hyperplane Linear SVM classification relies on is marginally more conservative than the sigmoid curve employed by Logistic regression.

4.3.1 Feature Importance & SHAPs Analysis

Below the VIMs scores for the Linear SVM model are rendered:

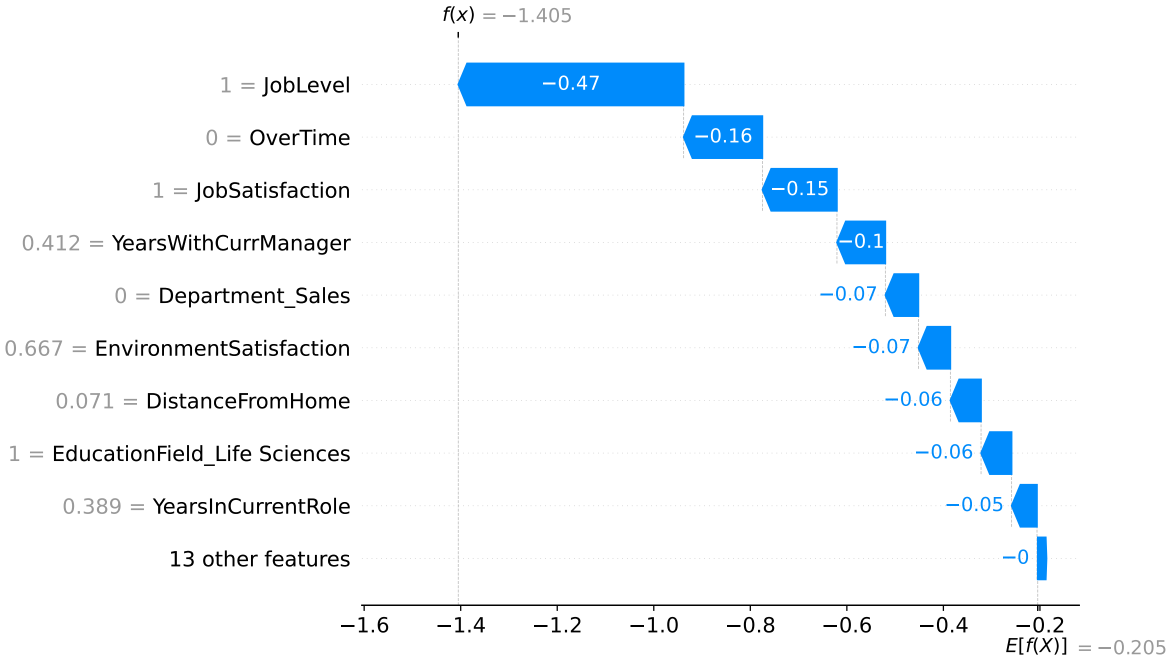


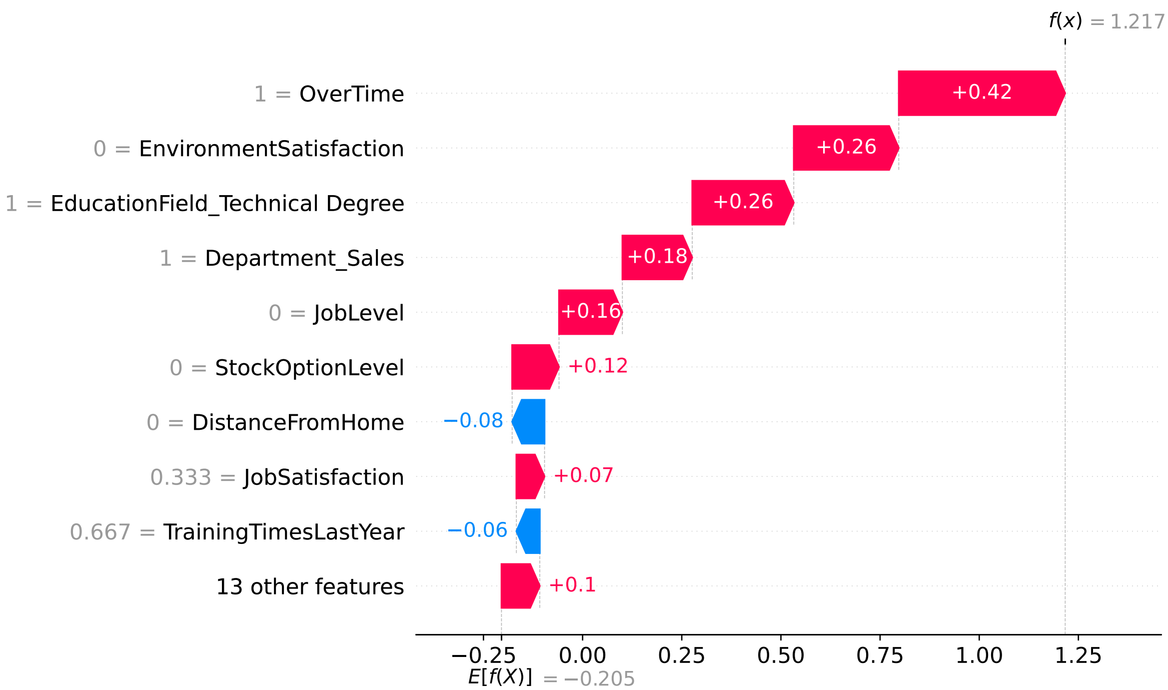

```
-> Added 22 features from Linear SVM (Tuned) to global tracker.
```

The striking similarity of the above plot with the Feature Importance graph produced by Model B offers us more in the way of convergent validity that Years in the company and Overtime are the key drivers of attrition. We can infer this convergence because Models B & C operate on two completely different mathematical frameworks (Probabilistic log odds vs Geometric Hyperplanes)

We now turn to SHAPs for a more localized view, however a different logic for the selection of edge cases is necessary due to the particularities of Linear SVM. Therefore we will select the points furthest away from the decision boundary on either the retained or terminated end. (i.e the employee furthest in the “safe” or “retention” zone is the lowest risk employee and the employee furthest in the “termination” or “danger” zone is the highst risk employee)

```
SVM - Furthest 'Safe' Point Index: 140 (Score: -1.4050)
SVM - Furthest 'Risk' Point Index: 47 (Score: 1.2167)

SHAP Explanation for SVM's Safest Employee (Index 140):

SHAP Explanation for SVM's Riskiest Employee (Index 47):
```





With the above plots, we now have consensus between Models A, B and C that employees 140 and 47 are the least and most risky respectively from the standpoint of turnover.

However, we also observe a critical divergence between Models B & C in the case of the highest risk employee (Employee #47), the Regualarized Logistic Regression Model saw much larger retention factors (long blue bars) indicating the model was actively wieghing Seniority (`JobLevel`) against burnout, viewing the employee as conflicted between loyalty and stress.

In Contrast, Model C significantly shrank these factors which might point to Linear SVM treating risk factors like `OverTime` (red bars) as non-negotiable violations. As mathematically, once an employee crosses the geometric threshold defined by these variables their positive attributes were taxed of their compensatory power as the SVM prioritizes the rigid enforcement of the decision boundary over a balancing act between model features to arrive at a prediction.

4.4 Decision Tree Algorithm (Model D)

The hyperparameters of interest in the Decision tree model are:

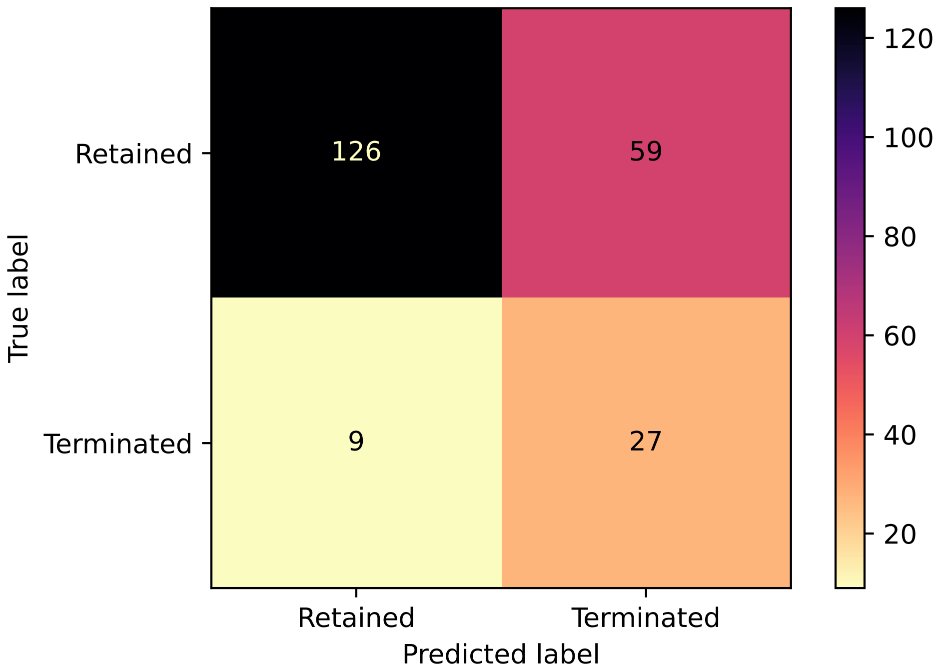
- **max depth**: This is the maximum depth (or generations) of a decision tree
- **min samples split**: This is the minimum number of samples to split a node
- **min samples leaf**: This is the minimum number of samples that must exist at the base of a decision tree to function as their own leaf or labeled outcome
- **criterion**: This is the loss function of the decion tree algorithm

Per Mantovani[4], values between 3 and 10 will be tested for `max_depth` and values between 20 and 50 will be grid searched for `min_samples_leaf`. As the Decision Tree Algorithm isn't subject to the deleterious effects of scaling, the unscaled raw featureset will be used for model training:

```
Optimal Parameters for the Decision Tree model are : {'min_samples_split': 2, 'min_samples_le
The test accuracy of the Decision Tree model with optimal parameters is : 0.44537733217638237
```

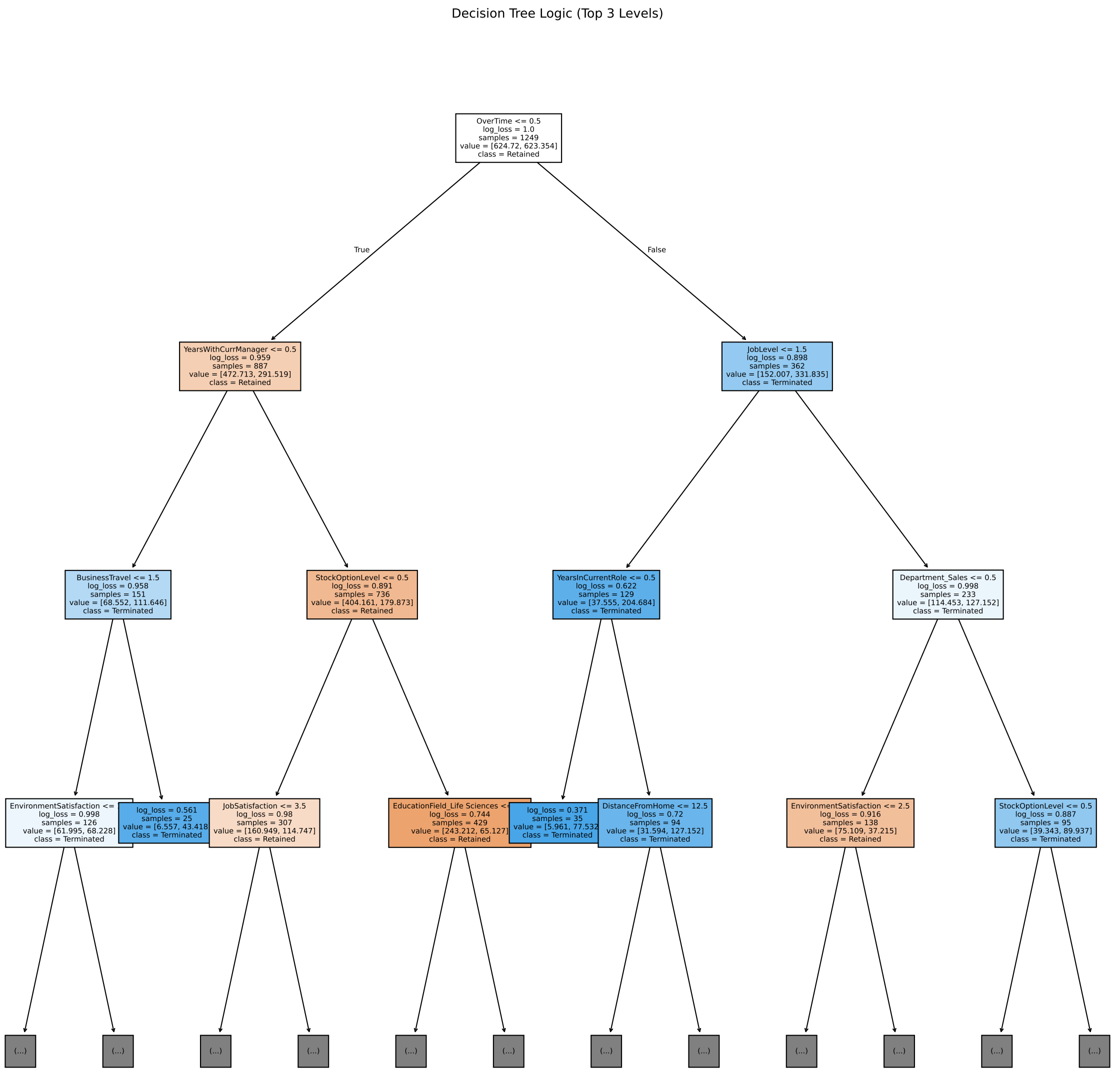
With optimal tuning criteria acheived we now turn to the construction of the decision tree and its classification report:

```
The confusion matrix for the Decision Tree model is :
None
The classification reprort for the Decision Tree model is :
              precision    recall  f1-score   support

           0       0.93      0.68      0.79       185
           1       0.31      0.75      0.44        36

    accuracy                           0.69       221
   macro avg       0.62      0.72      0.62       221
weighted avg       0.83      0.69      0.73       221
```



Here we observe the highest recall among Models A-D and comparable accuracy to Model B, we will construct a rendering of the decision tree itself for more insights

Below the top 3 levels of the trained decision tree model are rendered in a diagram:



As the diagram above has been truncated to preserve space,a text based representation of the full tree for enhanced readibilty

Note however, that the leaf nodes display the sum of class weights rather than raw sample counts. Due to the aggressive reweighting (Class 1 ≈ 5 times Class 0) the Decision ree frequently classifies a node as Terminated even when Leavers are the numerical minority, ensuring high sensitivity to attrition risk.

```
|--- OverTime <= 0.50
|   |--- YearsWithCurrManager <= 0.50
```

```
|   |   |--- BusinessTravel <= 1.50
|   |   |   |--- EnvironmentSatisfaction <= 1.50
|   |   |   |   |--- weights: [11.33, 37.22] class: 1
|   |   |   |--- EnvironmentSatisfaction >  1.50
|   |   |   |   |--- StockOptionLevel <= 0.50
|   |   |   |   |   |--- weights: [24.44, 27.91] class: 1
|   |   |   |   |--- StockOptionLevel >  0.50
|   |   |   |   |   |--- weights: [26.23, 3.10] class: 0
|   |   |--- BusinessTravel >  1.50
|   |   |   |--- weights: [6.56, 43.42] class: 1
|   |--- YearsWithCurrManager >  0.50
|   |   |--- StockOptionLevel <= 0.50
|   |   |   |--- JobSatisfaction <= 3.50
|   |   |   |   |--- EnvironmentSatisfaction <= 3.50
|   |   |   |   |   |--- weights: [71.53, 96.14] class: 1
|   |   |   |   |--- EnvironmentSatisfaction >  3.50
|   |   |   |   |   |--- weights: [33.98, 9.30] class: 0
|   |   |   |--- JobSatisfaction >  3.50
|   |   |   |   |--- EnvironmentSatisfaction <= 3.50
|   |   |   |   |   |--- weights: [35.17, 9.30] class: 0
|   |   |   |   |--- EnvironmentSatisfaction >  3.50
|   |   |   |   |   |--- weights: [20.27, 0.00] class: 0
|   |   |--- StockOptionLevel >  0.50
|   |   |   |--- EducationField_Life Sciences <= 0.50
|   |   |   |   |--- JobSatisfaction <= 3.50
|   |   |   |   |   |--- weights: [94.18, 21.71] class: 0
|   |   |   |   |--- JobSatisfaction >  3.50
|   |   |   |   |   |--- weights: [42.92, 31.01] class: 0
|   |   |   |--- EducationField_Life Sciences >  0.50
|   |   |   |   |--- YearsAtCompany <= 7.50
|   |   |   |   |   |--- weights: [57.82, 0.00] class: 0
|   |   |   |   |--- YearsAtCompany >  7.50
|   |   |   |   |   |--- weights: [48.28, 12.41] class: 0
|--- OverTime >  0.50
|   |--- JobLevel <= 1.50
|   |   |--- YearsInCurrentRole <= 0.50
|   |   |   |--- weights: [5.96, 77.53] class: 1
|   |   |--- YearsInCurrentRole >  0.50
|   |   |   |--- DistanceFromHome <= 12.50
|   |   |   |   |--- EnvironmentSatisfaction <= 2.50
|   |   |   |   |   |--- weights: [7.15, 43.42] class: 1
|   |   |   |   |--- EnvironmentSatisfaction >  2.50
|   |   |   |   |   |--- weights: [20.27, 43.42] class: 1
```

```
|   |   |   |--- DistanceFromHome >  12.50
|   |   |   |   |--- weights: [4.17, 40.32] class: 1
|   |--- JobLevel >  1.50
|   |   |--- Department_Sales <= 0.50
|   |   |   |--- EnvironmentSatisfaction <= 2.50
|   |   |   |   |--- EducationField_Medical <= 0.50
|   |   |   |   |   |--- weights: [9.54, 18.61] class: 1
|   |   |   |   |--- EducationField_Medical >  0.50
|   |   |   |   |   |--- weights: [11.33, 6.20] class: 0
|   |   |   |--- EnvironmentSatisfaction >  2.50
|   |   |   |   |--- JobLevel <= 2.50
|   |   |   |   |   |--- weights: [28.61, 0.00] class: 0
|   |   |   |   |--- JobLevel >  2.50
|   |   |   |   |   |--- weights: [25.63, 12.41] class: 0
|   |   |--- Department_Sales >  0.50
|   |   |   |--- StockOptionLevel <= 0.50
|   |   |   |   |--- DistanceFromHome <= 5.50
|   |   |   |   |   |--- weights: [7.15, 24.81] class: 1
|   |   |   |   |--- DistanceFromHome >  5.50
|   |   |   |   |   |--- weights: [4.77, 40.32] class: 1
|   |   |   |--- StockOptionLevel >  0.50
|   |   |   |   |--- YearsWithCurrManager <= 3.50
|   |   |   |   |   |--- weights: [10.13, 21.71] class: 1
|   |   |   |   |--- YearsWithCurrManager >  3.50
|   |   |   |   |   |--- weights: [17.29, 3.10] class: 0
```

The above logic and diagram bring to light 3 logical paths of high attrition or "kill-zones":

- 1. Junior Burnout: Identified via the path (OverTime > 0.50 → JobLevel <= 1.50 → Class 1) and bearing a class weight sum of 77.53, this represents one of the largest concentrations of turnover in the model and implies that junior employees working overtime roles are prime candidates for attrition
- 2. Deficient Commissions/Incentive Structures: Identified via the two branches of the path (OverTime > 0.50 → JobLevel > 1.50 → Department_Sales > 0.50 → StockOptionLevel <= 0.50 → Class 1) and having a combined class weight sum of 65.13, this path logically implies that Senior Sales Staff working overtime without stock options are high risk attrition candidiates which underscores a possible deficiency in the incentive or reward structure for Sales Staff.
- 3. Poor Managerial Fit: We can observe this trend from the (OverTime <= 0.50 → YearsWithCurrManager <= 0.50 → BusinessTravel <= 1.50 → EnvironmentSatisfaction <= 1.50 → Class 1) path present with a class wieght magnitude of 37.22 and validates the popular refrain in HR circles that employees leave bad managers not companies but only when the environmental situation is also poor

Unlike the linear models which prioritized Tenure, the Decision Tree identified `OverTime` as the root cause of attrition which served to split the population into some of the risk profiles deliniated above.

This structural difference explains why the Tree achieved the highest Recall (0.75) of all models so far as it captures the junior burnout segment that three prior linear models smoothed over.

4.5 kNN Model (Model E)

The hyperparameter of importance for the KNN algorithm is the `n_neighors` parameter which represents the number of neighbors factored into the prediction labels of the test set.

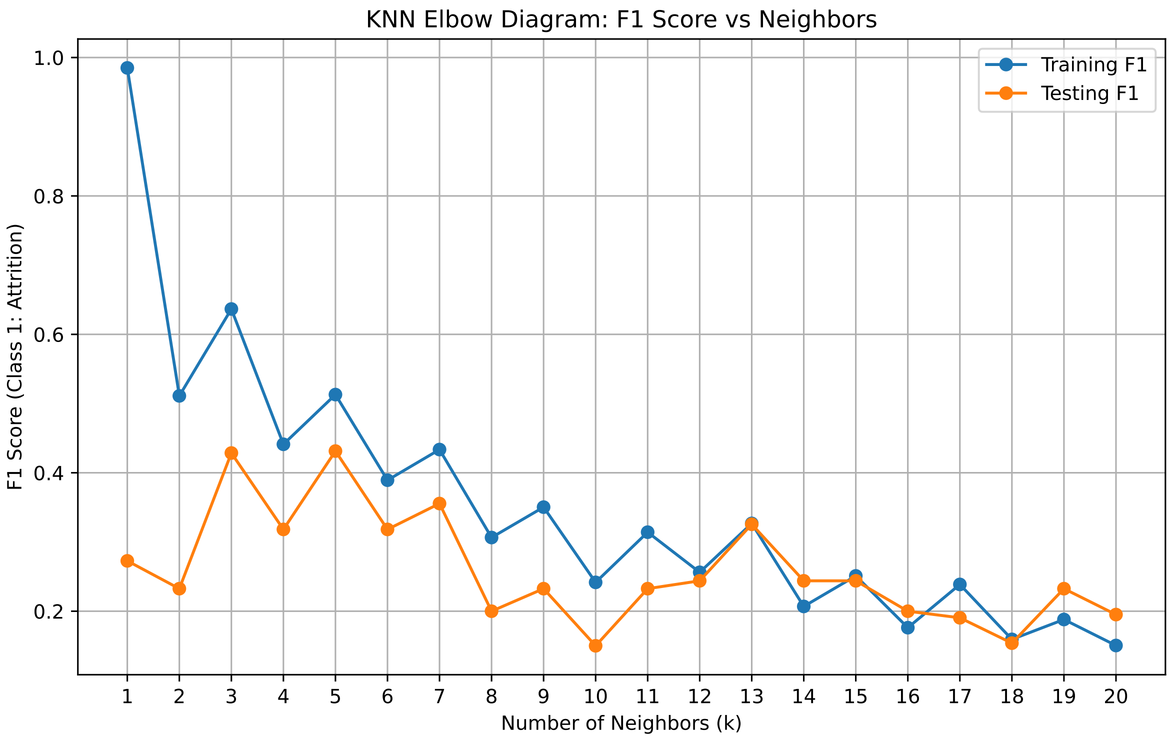
To arrive at the optimal number of neighbors we will use the elbow method as outlined in Section 2. Prior to this however, some feature engineering will be performed to reduce the number of features the KNN will have to compute distances from.

Since there is an abundance of categorical variables in this dataset, we will perform feature selection by means of univariate chi-squared tests on each feature which is made possible by sklearn's `SelectKBest` function which selects the best features on the basis of their ANOVA-F values (by determining for any given feature if the average value of employees who stay varies significantly from employees that leave):

```
kNN will now use only these 10 features:
['JobSatisfaction' 'StockOptionLevel' 'YearsAtCompany' 'JobLevel'
 'YearsWithCurrManager' 'YearsInCurrentRole' 'OverTime' 'BusinessTravel'
 'EnvironmentSatisfaction' 'Department_Research & Development']
```

We see here that the selected features are consistent with the features the other 4 models prioritize ('EnvironmentSatisfaction', 'JobLevel' 'OverTime' etc.) and will proceed to building the Elbow diagram to obtain the optimal number of neighbhors.

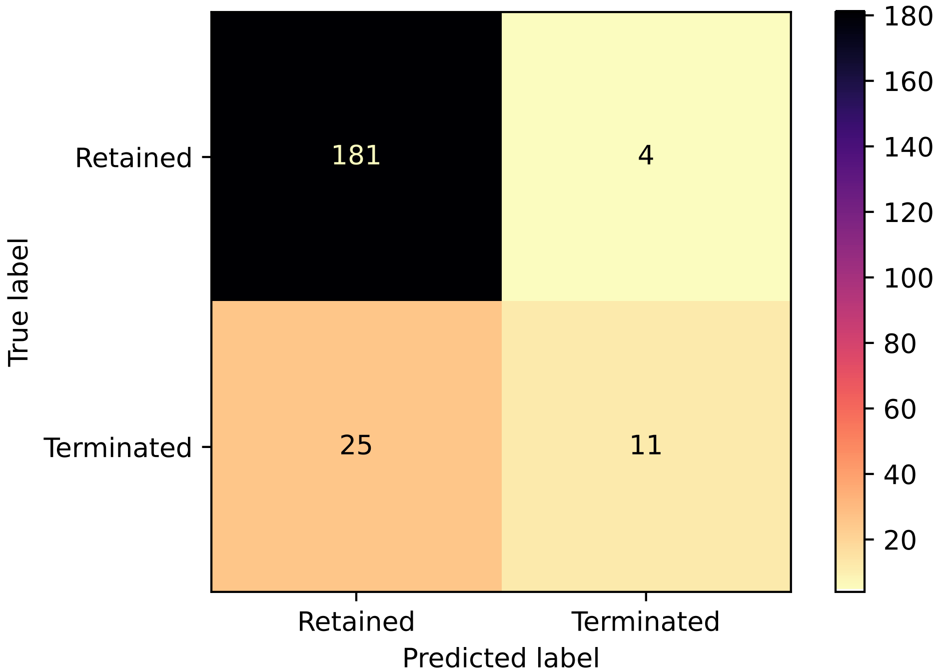
Elbow diagrams are calculated by looping over a range of numbers (1 to 10 in our case to test for k = 1 through k = 10) and obtaining f1 scores for models constructed with eached looped value of k:




```
Peak Performance: k=5 with Test F1 Score of 0.4314
```

Having now arrived at the most efficient number of neighbors, the kNN model with k = 5 is constructed and the Classification Summary is prepared:

```
The confusion matrix for the kNN Model (n = 5) model is :
None
The classification reprort for the kNN Model (n = 5) model is :
              precision    recall  f1-score   support

           0       0.88      0.98      0.93       185
           1       0.73      0.31      0.43        36

    accuracy                           0.87       221
   macro avg       0.81      0.64      0.68       221
weighted avg       0.85      0.87      0.85       221
```



With the results for kNN resembling Model A (0.35 recall vs 0.31 recall) and having comparable accuracy (0.82 vs 0.85), this observation points to less sophisticated models that are untuned and unweighted for class imbalance being unsuited to the nuance required to flag subtle cases of turnover trends.

5 Results

To arrive at a comprehensive overview of the analyis performed thus far, we aggregate the metrics of the classification reports to arrive at the best performing model by ranking them by positive class recall (called 'Attrition Recall' in the table):

```
                                 model name  Retention Recall  \
model rank
1.0                           Decision Tree              0.68
2.5         Regularized Logistic Regression              0.74
2.5                       Linear SVM (Tuned)             0.75
4.0                       kNN Model (n = 5)              0.98
5.0            Standard Logistic Regression              0.98

            Attrition Recall  f1 score
model rank
```

```
1.0                       0.75      0.73
2.5                       0.69      0.76
2.5                       0.69      0.77
4.0                       0.31      0.85
5.0                       0.22      0.83
```

From the summary above the triumph of regularization in reigning in the eccentricities of HR data is observable with Decision Tree and Tuned SVM and Logistic Regrssion models breaching the 50% threshold for Attrition recall, The Decision Tree's status as the best performing model of the ensemble can be attributed to its robustness in handling non-linear trends that the other models are not configured to do.

For Further insights we will consult the MARS framework and list the top 5 features favored by each model in a MARS context with labels appended to them based on each of the 4 MARS prongs:

```
=== TOP 5 PREDICTORS BY MODEL (With MARS Category) ===
Model    Decision Tree (Mantovani)        Linear SVM (Tuned)  \
Rank
1                     OverTime (R)        YearsAtCompany (A)
2                     JobLevel (A)  YearsWithCurrManager (S)
3             StockOptionLevel (M)              JobLevel (A)
4      EnvironmentSatisfaction (M)              OverTime (R)
5         YearsWithCurrManager (S)        BusinessTravel (S)

Model Regularized Logistic Regression Standard Logistic Regression  \
Rank
1                  YearsAtCompany (A)                 JobLevel (A)
2            YearsWithCurrManager (S)                 OverTime (R)
3                        JobLevel (A)     YearsWithCurrManager (S)
4              YearsInCurrentRole (R)         StockOptionLevel (M)
5                        OverTime (R)           BusinessTravel (S)

Model              kNN (k=5)
Rank
1               OverTime (R)
2               JobLevel (A)
3      TotalWorkingYears (A)
4         YearsAtCompany (A)
5       StockOptionLevel (M)
```

From our compilation above we now understand the following:

- 1. Burnout vs Stagnation Split: The non linear models (Decision Tree & kNN) have identified Role (R) stress (`OverTime`) as the top predictor pointing to insurmountable workloads leading to burnout. In contrast, the Regularized Linear Models (Ridge & SVM) identified Ability (A) (`YearsAtCompany`) as most influential predictor which indicates that the inertia of tenure overpowers even the `OverTime` variable in driving turnover trends
- 2. Managerial Support: The Situational (S) factor `YearsWithCurrManager` appears in the top 5 for four out of five models with kNN being the exception. It also features as the second most influential predictor for Regularized Logistic Regression & SVM. This affirms conventional wisdom in the HR space that confirms that managerial stability acts as a critical buffer against turnover.
- 3. Stock Options as a Silent Motivator: While Motivation (M) features were the least represented category as they appeared in the Top 5 only sporadically, their apparance in untuned models like kNN (Model E) and Standard Logistic Regression (Model A) suggests that while financial incentives matter, they are often overshadowed by the structural forces of Role (`OverTime`) and Ability (`JobLevel`)
- 4. Universal Attition Risk Factors: Every single model included both Role (OverTime) and Ability (JobLevel or YearsAtCompany) in its top 5 predictors despite the plurality in Algorithmic differences among them. This consensus validates the core hypothesis of the MARS framework for this dataset: Attrition is fundamentally a tugof war between the demands of the role (R) and the status of the employee (A) and models that fails to account for both those pillars risks statistical incompleteness.

5 References

[1] W. Wu and S. Fukui, "Using human resources data to predict turnover of community mental health employees: Prediction and interpretation of machine learning methods," *International Journal of Mental Health Nursing*, 2024, doi: 10.1111/inm.13386.

[2] Academy to Innovate HR, "What is HR analytics? All you need to know to get started." 2024. Available: https://www.aihr.com/blog/what-is-hr-analytics/

[3] P. M. Usha and N. V. Balaji, "A comparative study on machine learning algorithms for employee attrition prediction," *IOP Conference Series: Materials Science and Engineering*, vol. 1085, no. 1, p. 012029, 2021, doi: 10.1088/1757-899X/1085/1/012029.

[4] R. G. Mantovani, T. Horváth, R. Cerri, J. Vanschoren, and A. C. P. L. F. de Carvalho, "Hyper-parameter tuning of a decision tree induction algorithm," in *2016 5th brazilian conference on intelligent systems (BRACIS)*, IEEE, 2016, pp. 37–42. doi: 10.1109/BRACIS.2016.015.